

JAVA WITH DSA-DAY 11

PRINT THE LEADER ELEMENTS IN AN ARRAY (AN ELEMENT IS A LEADER IF IT IS GREATER THAN OR EQUAL TO ALL ELEMENTS TO ITS RIGHT):

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the size of the array: ");

        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.println("Enter the array elements:");

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

        }

        int max = arr[n - 1];

        ArrayList<Integer> leaders = new ArrayList<>();

        leaders.add(max);

        for (int i = n - 2; i >= 0; i--) {

            if (arr[i] >= max) {

                max = arr[i];

                leaders.add(arr[i]);

            }

        }

        Collections.reverse(leaders);

        System.out.println("Leader elements are:");

        for (int leader : leaders) {

            System.out.print(leader + " ");

        }

    }

}
```

OUTPUT:

Enter the size of the array: 6

Enter the array elements:

17

5

4

2

3

Leader elements are:

17 5 4 3

PRINT THE STRING WITHOUT DUPLICATES USING SET

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter String: ");

        String s = sc.nextLine();

        Set<Character> st=new LinkedHashSet<>();

        for(int i=0;i<s.length();i++){

            char ch=s.charAt(i);

            st.add(ch);

        }

        String res="";

        for(char ch:st){

            res+=ch;

        }

        System.out.println("String without duplicates: "+res);

    }

}
```

OUTPUT 1:

Enter String: Anukeerthana

String without duplicates: Anukertha

OUTPUT 2:

Enter String: PROGRAMMING

String without duplicates: PROGAMIN

UNION AND INTERSECTION OF TWO ARRAYS

```
import java.util.*;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter size of first array: ");  
        int n1 = sc.nextInt();  
        int arr1[] = new int[n1];  
        System.out.println("Enter first array elements:");  
        for (int i = 0; i < n1; i++) {  
            arr1[i] = sc.nextInt();  
        }  
        System.out.print("Enter size of second array: ");  
        int n2 = sc.nextInt();  
        int arr2[] = new int[n2];  
        System.out.println("Enter second array elements:");  
        for (int i = 0; i < n2; i++) {  
            arr2[i] = sc.nextInt();  
        }  
        // Union  
        Set<Integer> union = new LinkedHashSet<>();  
        for (int i = 0; i < n1; i++) {  
            union.add(arr1[i]);  
        }  
        for (int i = 0; i < n2; i++) {  
            union.add(arr2[i]);  
        }  
        System.out.println("Union: " + union);  
        // Intersection  
        Set<Integer> set1 = new LinkedHashSet<>();  
        Set<Integer> intersection = new LinkedHashSet<>();  
        for (int i = 0; i < n1; i++) {  
            set1.add(arr1[i]);  
        }  
        for (int i = 0; i < n2; i++) {
```

```

        if (set1.contains(arr2[i])) {
            intersection.add(arr2[i]);
        }
    }
    System.out.println("Intersection: " + intersection);
}
}

```

OUTPUT:

Enter size of first array: 4

Enter first array elements:

21

56

12

94

Enter size of second array: 5

Enter second array elements:

12

90

28

63

56

Union: [21, 56, 12, 94, 90, 28, 63]

Intersection: [12, 56]

In strings:

```
import java.util.*;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter first string: ");
```

```
        String s1 = sc.nextLine();
```

```
        System.out.print("Enter second string: ");
```

```
        String s2 = sc.nextLine();
```

```
        // Union
```

```
        Set<Character> union = new LinkedHashSet<>();
```

```

for (int i = 0; i < s1.length(); i++) {
    union.add(s1.charAt(i));
}

for (int i = 0; i < s2.length(); i++) {
    union.add(s2.charAt(i));
}

// Intersection

Set<Character> set1 = new LinkedHashSet<>();
Set<Character> intersection = new LinkedHashSet<>();

for (int i = 0; i < s1.length(); i++) {
    set1.add(s1.charAt(i));
}

for (int i = 0; i < s2.length(); i++) {
    if (set1.contains(s2.charAt(i))) {
        intersection.add(s2.charAt(i));
    }
}

System.out.println("Union: " + union);
System.out.println("Intersection: " + intersection);
}
}

```

OUTPUT:

Enter first string: hello

Enter second string: world

Union: [h, e, l, o, w, r, d]

Intersection: [o, l]

CHECK WHETHER AN ARRAY CONTAINS ANY DUPLICATE ELEMENT WITHIN A SLIDING WINDOW OF SIZE K USING A HASHSET

```
import java.util.*;
```

```

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");

        int n = sc.nextInt();
    }
}

```

```

int[] arr = new int[n];
System.out.println("Enter elements:");
for (int i = 0; i < n; i++) {
    arr[i] = sc.nextInt();
}
System.out.print("Enter k: ");
int k = sc.nextInt();
Set<Integer> set = new HashSet<>();
for (int i = 0; i < n; i++) {
    if (set.contains(arr[i])) {
        System.out.println("True");
        return;
    }
    set.add(arr[i]);
    if (set.size() > k) {
        set.remove(arr[i - k]);
    }
}
System.out.println("False");
}
}

```

OUTPUT:

Enter number of elements: 6

Enter elements:

1 2 4 5 2 6

Enter k: 3

True